

Quantum Computing Assignment - 2

Total Marks: 20

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Consult and study the following materials before attempting the assignment:

1. Class lecture notes
2. IBM Quantum Learning Module:
<https://quantum.cloud.ibm.com/learning/en/courses/fundamentals-of-quantum-algorithms/quantum-query-algorithms/simon-algorithm>
3. Qiskit Textbook implementation:
<https://github.com/qiskit-community/qiskit-textbook/blob/main/content/ch-algorithms/simon.ipynb>

In this assignment you will implement Simon's Algorithm using Qiskit to determine a hidden binary string s .

Simon's algorithm solves the following problem. A function $f: \{0,1\}^n \rightarrow \{0,1\}^n$ satisfies the Simon promise:

$$f(x) = f(x \oplus s)$$

for some secret string $s \neq 0$.

The goal is to determine the hidden string s . Each execution of the quantum circuit produces a bit string u satisfying: $u \cdot s = 0 \pmod{2}$

These measurement outcomes can then be used to construct a system of linear equations whose solution reveals the secret string. Classically this problem requires an exponential number of oracle queries, while Simon's quantum algorithm determines s using $O(n)$ queries.

Part A — Understanding the Reference Implementation (4 Marks)

Study the references:

The Quantum cloud and the notebook demonstrate Simon's algorithm for the secret string

$s = 011$

Tasks

- Run the notebook in Qiskit and verify that the circuit executes correctly.
- Explain in your own words the role of:
 - the Hadamard gates
 - the oracle
 - the second Hadamard layer

– the measurement

- Briefly describe why the measurement outputs satisfy: $u \cdot s = 0$

Part B — Implement Simon's Algorithm for a New Secret String (10 Marks)

Now modify the implementation to test Simon's algorithm for a **new secret string of length 5 bits**. Choose any non-trivial 5-bit string of your choice, for example $s = 10110$

Tasks

- Implement the Simon oracle corresponding to your chosen secret string.
- Construct the full Simon quantum circuit in Qiskit.
- Execute the circuit using the Qiskit simulator with at least 2000 shots.
- Record the measurement outcomes obtained from the first register.

Your output should produce several bit strings u .

Exclude the trivial outcome 00000 from further analysis.

Part C — Classical Post-Processing (6 Marks)

Each measured bit string u satisfies: $u \cdot s = 0 \pmod{2}$

Use the measurement outcomes to construct a system of linear equations.

For example,

$$u_1 \cdot s = 0$$

$$u_2 \cdot s = 0$$

$$u_3 \cdot s = 0$$

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Tasks

- Extract the measured bit strings u from the experiment in Part- B
- Using any programming tool of your choice, for example: Python; MATLAB; Mathematica; any other numerical tool of your choice: Construct the matrix equation: $Us = 0 \pmod{2}$ and solve this system of equations
- Determine the secret string s from the equations.

Submit:

GitHub Repository

- A Qiskit notebook (.ipynb) implementing the reference Simon's algorithm example for the 3-bit secret string $s = 011$ (based on the provided materials).
- A Qiskit notebook (.ipynb) implementing Simon's algorithm for a 5-bit secret string of your choice.
- The classical code/script (Python, MATLAB, etc.) used to solve the system of equations $Us = 0 \pmod{2}$.

- A short README.md stating:
 - the chosen 5-bit secret string,
 - how to run the notebooks,
 - the recovered value of s .

PDF Report / Notes

- The Tasks in Part -A of the assignment
- List of the measured strings u obtained from the circuit (excluding the trivial outcome 00000) and the corresponding system of linear equations $u \cdot s = 0 \pmod{2}$
- Demonstration of the classical solution of the system $Us = 0$, verifying that the recovered string matches the original secret string